SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

Name	SABRINA HENG WEI QI
IC No. / Matric No.	040511040204 / A23CS0265
Year / Program	2/SECPH
Section	02
Lecturer Name	DR. SEAH CHOON SEN
Case Study	Member Management Module
System Name	KADA ESERVE
Group Name	TECHAWAY

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

SQL Statement (DDL & DML)

Guest Table:

CREATE TABLE Guest (
id INT PRIMARY KEY,
password VARCHAR (255) NOT NULL,
name VARCHAR (255) NOT NULL,
email VARCHAR (255) NOT NULL
);

←T→	id	name	email	email_verified_at	password

Insert Data into Guest Table
INSERT INTO Guest (id, password, name, email)
VALUES (1, ‘admin123’, ‘Admin’, ‘admin@example.com’);

←T→	id	name	email	email_verified_at	password
Edit Copy Delete	1	Admin	admin@example.com	NULL	$2y$12$jw75hUcFtyVrdQnsWcD2BulQppbahvTY3Lk5PAaTtTb...

Delete Guest Record
DELETE FROM Guest
WHERE id = 1;

←T→	id	name	email	email_verified_at	password

Member Table:

Insert Data into Member table
INSERT INTO Member (
id, created_at, updated_at, no_anggota, name, email, ic, phone, address, city, postcode, state, gender, DOB, agama, bangsa, no_pf, salary, office_address, office_city, office_postcode, office_state, guest_id, status
)
VALUES
(101, NOW(), NOW(), 'A001', 'John Doe', 'johndoe@gmail.com', '910101-01-1234', '0123456789', '123 Main St', 'City A', '12345', 'State A', 'M', '1991-01-01', 'Buddha', 'Cina', 121, 5000, '123 Main St', 'City A', '12345', 'State A', 1, 'pending'),
(102, NOW(), NOW(), 'A002', 'Jane Smith', 'janesmith@gmail.com', '920202-02-2345', '0198765432', '456 Elm St', 'City B', '23456', 'State B', 'F', '1992-02-02', 'Buddha', 'Cina', 122, 5200, '123 Main St', 'City A', '12345', 'State A', 2, 'pending');

id	created_at	updated_at	no_anggota	name	email	ic	phone	address	city	postcode	state	gender	DOB	agama	bangsa	no_pf	salary	office_address	office_city	office_postco
101	2025-01-27 03:11:45	2025-01-27 03:11:45	A001	John Doe	johndoe@gmail.com	910101-01-1234	0123456789	123 Main St	City A	12345	State A	M	1991-01-01	Buddha	Cina	121	5000	123 Main St	City A	12345
102	2025-01-27 03:11:45	2025-01-27 03:11:45	A002	Jane Smith	janesmith@gmail.com	920202-02-2345	0198765432	456 Elm St	City B	23456	State B	F	1992-02-02	Buddha	Cina	122	5200	123 Main St	City A	12345

Update Status of a Member
UPDATE Member
SET status = ‘approved’
WHERE id = 101;

no_pf	salary	office_address	office_city	office_postcode	office_state	guest_id	status
121	5000	123 Main St	City A	12345	State A	1	approved
122	5200	123 Main St	City A	12345	State A	2	pending

Admin Table:
Create Admin Table
CREATE TABLE Admin (
    id INT PRIMARY KEY,
    password VARCHAR (255) NOT NULL,
    name VARCHAR (255) NOT NULL,
    email VARCHAR (255) NOT NULL
);

←T→	id	name	email	email_verified_at	password

Insert Data into Admin Table
INSERT INTO Admin (id, password, name, email)
VALUES (1, ‘admin123’, ‘Admin’, ‘admin@example.com’);

←T→	id	name	email	email_verified_at	password
Edit Copy Delete	1	Admin	admin@example.com	NULL	$2y$12$jw75hUcFtyVrdQnsWcD2BulQppbahvTY3Lk5PAaTtTb...

Family Table:
Create Family Table
CREATE TABLE Family (
    id INT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    ic VARCHAR(14) NOT NULL,
    relationship VARCHAR(50) NOT NULL,
    no_anggota VARCHAR (50) NOT NULL
);

id	created_at	updated_at	relationship	name	ic	no_anggota

Insert Data into Family Table
INSERT INTO Family (ID, name, ic, relationship,no_anggota)
VALUES (1, 'David Doe', '910101-01-5555', 'Father', ‘101’),
(2, 'Emily Doe', '920202-02-6666', 'Mother', ‘102’);

←T→	id	created_at	updated_at	relationship	name	ic	no_a
Edit Copy Delete	1	NULL	NULL	Father	David Doe	910101-01-5555	
Edit Copy Delete	2	NULL	NULL	Mother	Emily Doe	920202-02-6666	

StaffInfo Table:
Insert Data into working_info Table
INSERT INTO working_info (id, jawatan, gred, no_pf,salary,no_anggota)
VALUES (1, 'Manager', 'G5','123',5000,'101'),
(2, 'Clerk', 'G2','456',5200,'102');

←T→	id	created_at	updated_at	jawatan	gred	no_pf	salary	no_a
Edit Copy Delete	1	NULL	NULL	Manager	G5	123	5000	
Edit Copy Delete	2	NULL	NULL	Clerk	G2	456	5200	

Savings Table:
Insert Data into Savings Table
INSERT INTO Savings (id, entrance_fee, share_capital, subscription_capital, member_deposit, welfare_fund, fixed_savings, total_amount, no_anggota)
VALUES
(1, 100.00, 1000.00, 200.00, 500.00, 50.00, 3000.00, 6000.00, 101),
(2, 50.00, 500.00, 100.00, 300.00, 25.00, 1500.00, 3500.00, 102);

←T→	id	created_at	updated_at	entrance_fee	share_capital	subscription_capital	member_deposit	welfare_fund	fixed_savings	total_amount	no_anggota
Edit Copy Delete	1	NULL	NULL	100.00	1000.00	200.00	500.00	50.00	3000.00	6000.00	101
Edit Copy Delete	2	NULL	NULL	50.00	500.00	100.00	300.00	25.00	1500.00	3500.00	102

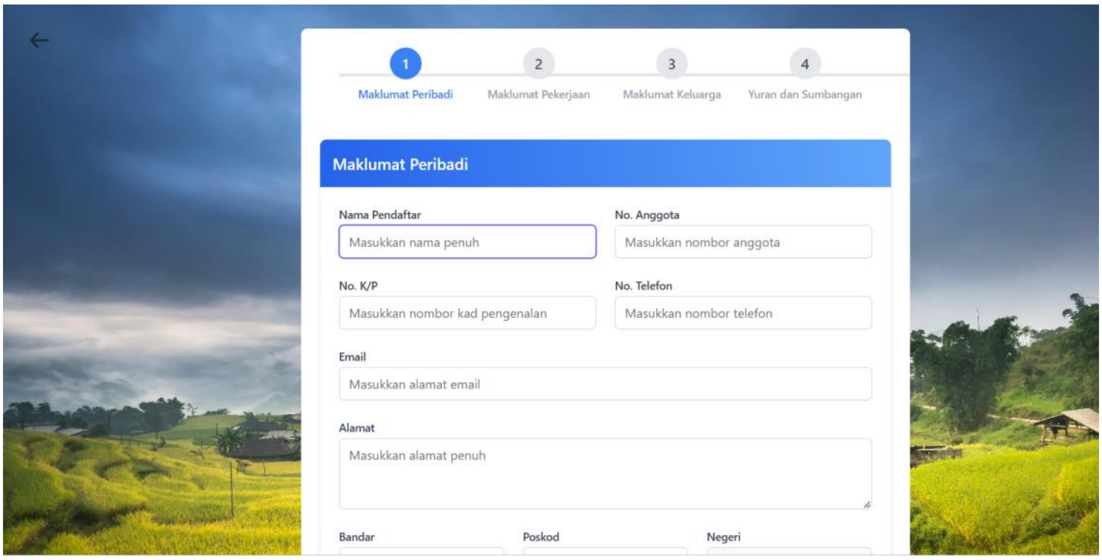
Join Queries
Query Member and Guest
SELECT Member.id AS member_id, Member.name, Guest.name AS guestUsername, Guest.email AS guestEmail
FROM Member
INNER JOIN Guest ON Member.guest_id = Guest.id;

Query Member and Family
SELECT Member.member_id, Member.name, Guest.guestUsername, Guest.guestEmail
FROM Member
INNER JOIN Guest ON Member.guest_id = Guest.guest_id;

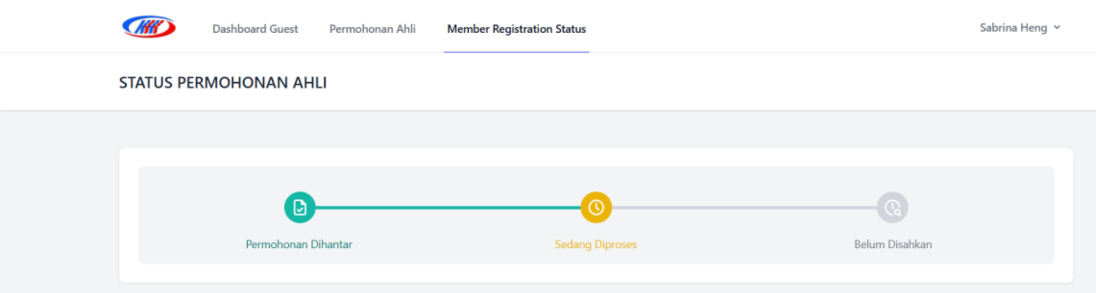
Query Member and Savings

SELECT Member.member_id, Member.name, Savings.savingAmount, Savings.dividendRate
FROM Member
INNER JOIN Savings ON Member.member_id = Savings.member_id;

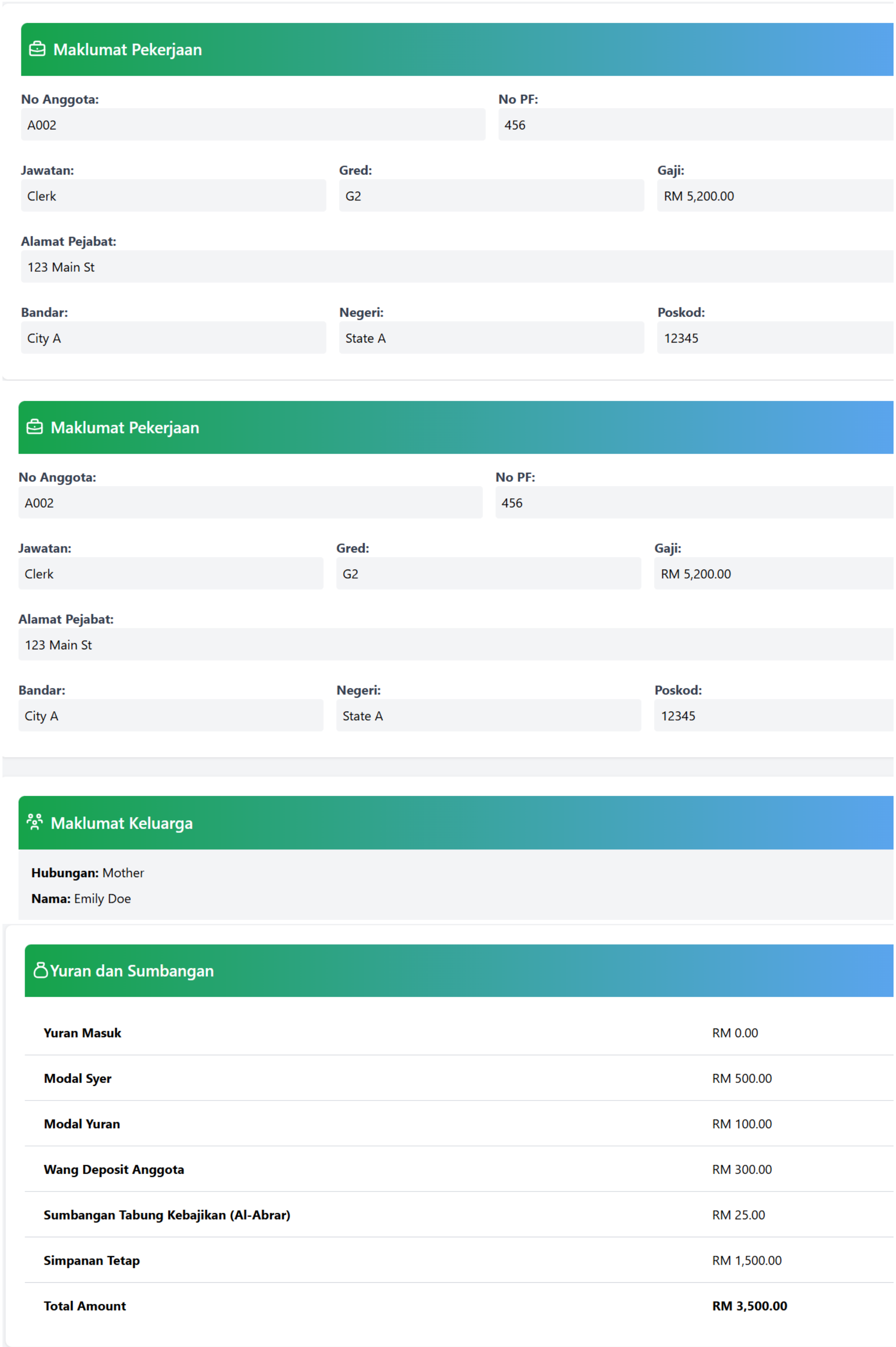
Interface Implementation



The interface for getting member registration information for applying to become a member. These information entered will be save in several tables that created such as member table, family table and working_info table.



After submitting the member registration form, all of the information will be store into the database. Then we will saw some data will be retrieve out like the status of the member registration. The status will be shown by retrieving out the status attribute from the member table.



Otherwise, in the member registration status page not only showing the application status. It will also show the member registration form information that be inserted by the guest. This part is retrieving all of the information have been stored in different database table such as member table, family table, working_info table and also savings table.

SECTION D (CONCLUSION AND REFLECTION)

4. Conclusion

1

The Member Management Module of the KADA eServe system has successfully transformed the manual, paper-based member registration process into a streamlined, digital workflow. It has reduced errors from manual data entry and made the onboarding process easier for both members and admin. The module securely stores member data in a centralized database and integrates well with other parts of the system, like savings and loan management. Additionally, features like role-based access control ensure that only authorized users can access sensitive member information which helps to improve security.

However, there are still ways to improve the module. The user interface couble be made simpler and more user-friendly, with features like progress indicators and tooltips to guide users for registering as a member. Adding support for multiple languages such as Bahasa Melayu and English would also help more members use it easily.

By working on these improvements, the Member Management Module will be evolve into a more user-friendly system which able to ensure a long-term use for the company.

4. Reflection

In this project, our team involved in developing the KADA eServe system to replace the KADA's paper-based mode for member registration and loan application process. There is also involve others feature such as viewing reports and have a savings account for each member of KADA. This system is develop for the Koperasi KADA for making their process more efficient and secure. During this project, my main responsibility was designing and implementing the Member Management Module, which digitized member registration, automated data validation and securely stored information in a centralized database. Otherwise, I am also the leader of Database subject, which having a role of lead and guiding my teammates to proceed for all the documentation about database. Our team worked collaboratively, with each member assigned specific roles like loan application, reporting, savings and more.

Throughout the project, I felt both excited and challenged. While implementing database functionality and SQL queries, I faced issues such as data inconsistency and integration delays. However, successfully completing features like real-time validation and data storage gave me a strong sense of achievement. For documentation work has also make me worries so much as we must make this report like the system and other's subject report. I am glad that I am having help from my lecturer, Dr Seah Choon Sen and my teammate by giving advice for ways to solve problems. The experience also improved my technical skills and highlighted the importance of teamwork and clear communication but also making me to use a long time for completing works.

In conclusion, this project taught me the value of detailed planning and proactive problem-solving. Moreover, it also teaches me about ways to communicate with teammates and learn to guide a team. I am happy that my teammates trust me so much and always gives advice for me when facing problems. Currently, we have solved most of the questions and getting to the end of the system development.

To improve in the future, I aim to allocate more time for testing and debugging, maintain a better documentation and explore tools to enhance efficiency. By learning from these challenges and successes, I am confident that I will contribute more effectively to similar projects in the future. At the end, saying thank you again to my lecturer and the best teammates.